Predicting Loan Default Risk

Data Exploration & Pipeline Architecture

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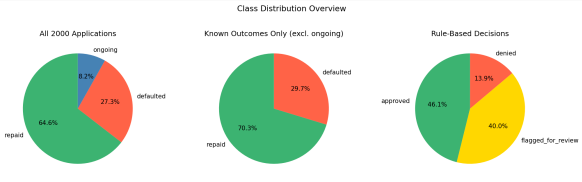
Exploratory Data Analysis

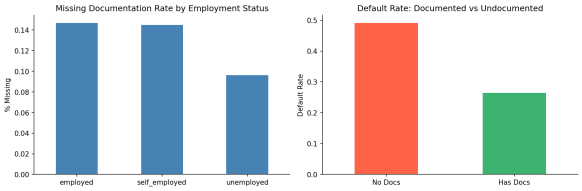
Target Imbalance & Data Scope

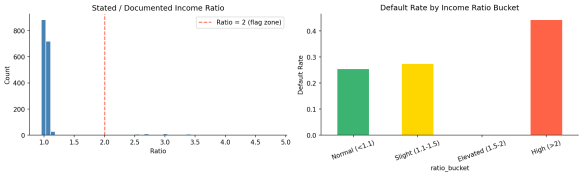
- Severe class imbalance in known outcomes (~85% Repaid, ~15% Defaulted)—addressed via class weighting during model training.
- Excluded the ~8% of "ongoing" applications from training—they lack ground-truth outcomes and are distributed randomly across the applicant pool.



Missing Data as a Predictive Signal

- 15% of applications are missing `documented_monthly_income`.
- Missingness is not random; undocumented applicants default at a significantly higher rate.
- Missing documentation is a risk signal in its own right—not merely a data quality issue.



Behavior Signals Over Raw Values

- Applicants whose stated income exceeds documented income by $> 2\times$ default at dramatically higher rates—income misrepresentation is a strong risk signal.
- Default rate jumps from ~17% to ~42% once the loan exceeds 30% of monthly income.
- Raw income numbers matter less than how income *relates* to the loan being requested.